CHAPTER

06

Electromagnetism and Electronics

Electricity and magnetism run almost every device you use: phones, computers, motors, lights, the power grid. This chapter builds the subject step by step: **charges** and the forces and fields they create; **potential** and stored energy; **current** and circuits; **magnetic fields** and **electromagnetic induction** (how generators and transformers work); and finally **semiconductors**, **diodes** and **transistors**, the building blocks of modern electronics.

LESSONS IN THIS CHAPTER

6.1	Electric Charge, Coulomb's Law and Electric Field
6.2	Electric Potential and Capacitance
6.3	Current, Resistance, Ohm's Law and Electric Power
6.4	Resistor Combinations, EMF and Kirchhoff's Rules
6.5	Magnetic Field, Magnetic Force and Magnetic Flux
6.6	Electromagnetic Induction
6.7	Conductors, Insulators and Semiconductors
6.8	Diodes and Transistors

Freshman A+ · General Physics (Phys 1011)

CHAPTER 6 · LESSON 6.1

Electric Charge, Coulomb's Law and Electric Field

AFTER THIS LESSON YOU CAN

- Describe the properties of electric charge
- Distinguish conductors and insulators and explain charging methods
- Apply Coulomb's law, including vector superposition
- Calculate and draw electric fields

6.1.1 Electric charge

D PROPERTIES OF CHARGE

- There are **two kinds**: **positive** (protons) and **negative** (electrons). **Like charges repel; unlike charges attract.**
- Charge is **conserved**: it is never created or destroyed, only transferred (rubbing a glass rod with silk moves electrons from the glass to the silk).
- Charge is **quantized**: every charge is a whole-number multiple of the elementary charge $e = 1.60 \times 10^{-19}$ C: $q = ne$.
- SI unit: the **coulomb** (C). 1 µC = 10^{-6} C, 1 nC = 10^{-9} C.
- Electron: $-e$, $m_e = 9.11 \times 10^{-31}$ kg. Proton: $+e$, $m_p = 1.67 \times 10^{-27}$ kg.

★ CONDUCTORS AND INSULATORS

- **Conductors** (metals, salt water, the human body) have **free electrons** that move easily.
- **Insulators** (glass, rubber, plastic, dry wood) hold their electrons tightly.
- **Semiconductors** (silicon, germanium) are in between (lesson 6.7).

E WAYS TO CHARGE AN OBJECT

- **Friction:** rubbing transfers electrons.
- **Conduction (contact):** touching a charged body shares its charge (same sign).
- **Induction:** bringing a charged body **near** a conductor separates its charges; grounding then leaves the conductor with the **opposite** sign, without contact.

6.1.2 Coulomb's law

D COULOMB'S LAW

The electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. It acts along the line joining them.

$$F = k\frac{|q_1 q_2|}{r^2}, \qquad k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}, \qquad \varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2/(N \cdot m^2)}$$

- Same form as gravity ($1/r^2$), but **much stronger** and can **attract or repel**.

- Forces on the two charges are **equal and opposite** (Newton's third law), even if the charges are unequal.
- With several charges, the net force is the **vector sum** of the individual forces (**superposition**).

SCALING RULES

Change	Effect on F
double one charge	× 2
double both charges	× 4
double the distance	× ¼
halve the distance	× 4

WORKED EXAMPLE 1: TWO CHARGES

$q_1 = +3.0$ µC and $q_2 = -5.0$ µC are 30 cm apart. Find the force.

$F = (8.99 \times 10^9)\dfrac{(3.0 \times 10^{-6})(5.0 \times 10^{-6})}{(0.30)^2} = 1.5$ N, **attractive** (opposite signs).

WORKED EXAMPLE 2: ELECTRIC VS GRAVITATIONAL FORCE IN HYDROGEN

The electron and proton are 5.3×10^{-11} m apart.

$F_e = \dfrac{(8.99 \times 10^9)(1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} = 8.2 \times 10^{-8}$ N; $F_g = \dfrac{G m_e m_p}{r^2} = 3.6 \times 10^{-47}$ N.

The electric force is about 10^{39} times stronger, which is why gravity is irrelevant inside atoms.

6.1.3 The electric field

ELECTRIC FIELD

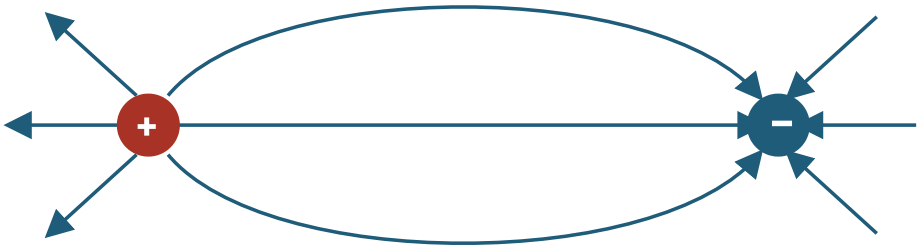
The electric force per unit **positive test charge** at a point:

$$\vec{E} = \frac{\vec{F}}{q_0} \qquad (\text{unit: N/C} = \text{V/m})$$

For a point charge: $E = k\dfrac{|q|}{r^2}$, pointing **away** from a positive charge and **toward** a negative charge. A charge q in a field feels $\vec{F} = q\vec{E}$ (a negative charge is pushed **opposite** to $\vec{E}$).

ELECTRIC FIELD LINES

- Start on **positive** charges and end on **negative** charges (or at infinity).
- The field is **tangent** to the lines; **closer lines mean a stronger field**.
- Field lines **never cross**.
- Between two parallel plates with opposite charges the field is **uniform**: $E = V/d$.



field lines run from + to –

Field lines of an electric dipole

WORKED EXAMPLE 3: WHERE IS THE FIELD ZERO?

Charges $+q$ and $+4q$ are a distance d apart. Where between them is $E = 0$?

At distance x from $+q$: $\frac{kq}{x^2} = \frac{k(4q)}{(d-x)^2} \Rightarrow d - x = 2x \Rightarrow x = \frac{d}{3}$ (closer to the **smaller** charge).

WORKED EXAMPLE 4: EQUILATERAL TRIANGLE (AAU FINAL 2014)

$q_1 = -q$ and $q_2 = +q$ sit at two corners of an equilateral triangle of side a. Find the field at the third corner.

Each charge gives a field of the same size $E_1 = kq/a^2$: from $+q$ it points **away**, and from $-q$ it points **toward** that charge. The two vectors are 120° apart, so the resultant has the same magnitude $E = kq/a^2$, directed **parallel to the line joining the charges, from the $+q$ side toward the $-q$ side**.

With q in µC and $a = 0.10$ m: $E = \frac{(9.0 \times 10^9)(q \times 10^{-6})}{0.010} = 9.0 \times 10^5\, q$ N/C.

CHARGE AND FIELD TRAPS

- The force on each of two charges is the **same size**, even if one charge is much larger.
- Coulomb's law uses the distance **squared**; convert cm to m and µC to C.
- The field at a point exists whether or not a charge is placed there.
- A field of E at 2 m becomes $E/4$ at **4 m**, not at 8 m.

Exam Practice

8 QUESTIONS · LESSON 6.1

1 Addis Ababa University · Final 2014

Three objects are brought together two at a time. A and B repel; B and C also repel. Which statement is true?

A) A and B have charges of opposite sign
B) A and C have charges of opposite sign
C) A, B and C all have charges of the same sign
D) B and C have charges of opposite sign

2 Addis Ababa University · Final 2014

How can you quadruple the electric force between two charged particles?

A) Double the distance and double the product of the charges
B) Halve the distance (or quadruple the product of the charges)
C) Double one charge and keep the distance the same
D) Double each charge and double the distance

3 Wolaita Sodo University · Final Exam 2014

An isolated point charge produces a field of magnitude E at 2 m. At what distance is the field $E/4$?

A) 1 m
B) 0.5 m
C) 2 m
D) 4 m

4 Model question

The charge on any object is always a whole-number multiple of 1.6×10^{-19} C. This property is called:

A) quantization of charge
B) conservation of charge
C) polarization
D) induction

5 Model question

How many electrons must be removed from a neutral object to give it a charge of $+3.2$ μC?

A) 2.0×10^{13}
B) 5.1×10^{-25}
C) 2.0×10^{19}
D) 3.2×10^{6}

6 Addis Ababa University · Final 2014

Point charges $q_1 = -q$ μC and $q_2 = +q$ μC are at two vertices of an equilateral triangle of side $a = 10$ cm. Find the magnitude and direction of the net electric field at the third vertex.

7 Serway / OpenStax-style problem

Three charges lie on the x-axis: $+2.0$ μC at $x = 0$, $+3.0$ μC at $x = 0.20$ m, and -4.0 μC at $x = 0.40$ m. Find the net force on the $+3.0$ μC charge.

8 Model question

Find the electric field 3.0 cm from a 2.0 nC point charge. What force would an electron feel there?

Answers & Explanations

1. **C**

A, B and C all have charges of the same sign

Repulsion means like charges: A has the same sign as B, and C has the same sign as B, so all three match.

2. **B**

Halve the distance (or quadruple the product of the charges)

$F \propto q_1 q_2 / r^2$. Halving r gives × 4; so does quadrupling $q_1 q_2$. A gives × ½, C gives × 2, D gives × 1.

3. **D**

4 m

$E \propto 1/r^2$: one quarter of the field means twice the distance, 4 m.

4. **A**

quantization of charge

Charge comes in discrete units of e: $q = ne$.

5. **A**

2.0×10^{13}

$n = q/e = 3.2 \times 10^{-6} / 1.6 \times 10^{-19} = 2.0 \times 10^{13}$.

6. **E = kq/a² = 9.0 × 10⁵ q N/C, parallel to the line joining the charges, pointing from the +q side toward the −q side**

Each field has magnitude kq/a^2; one points away from $+q$, the other toward $-q$; they are 120° apart, so the resultant equals either one in size and is parallel to the base.

7. **4.0 N in the +x direction**

From $+2.0$ µC (repulsive, pushes in $+x$): $F = \dfrac{(9 \times 10^9)(2 \times 10^{-6})(3 \times 10^{-6})}{0.04} = 1.35$ N.

From -4.0 µC (attractive, pulls in $+x$): $F = \dfrac{(9 \times 10^9)(4 \times 10^{-6})(3 \times 10^{-6})}{0.04} = 2.70$ N.

Net $= 4.05 \approx 4.0$ N in $+x$.

8. **E = 2.0 × 10⁴ N/C (away from the charge); F = 3.2 × 10⁻¹⁵ N toward the charge**

$E = kq/r^2 = (9 \times 10^9)(2 \times 10^{-9})/(0.030)^2 = 2.0 \times 10^4$ N/C. $F = eE = 3.2 \times 10^{-15}$ N, opposite to $\vec{E}$.

CHAPTER 6 · LESSON 6.2

Electric Potential and Capacitance

AFTER THIS LESSON YOU CAN

- Define electric potential energy, potential and potential difference
- Relate potential to work and to a uniform field
- Use the electron-volt
- Calculate capacitance, combinations and stored energy

6.2.1 Potential energy and potential

D ELECTRIC POTENTIAL

The **electric potential** V at a point is the **potential energy per unit charge** there:

$$V = \frac{U}{q} \qquad (\text{unit: volt, } 1\ \text{V} = 1\ \text{J/C})$$

For a point charge: $V = k\dfrac{q}{r}$ (a **scalar**, with the sign of q). The potential of several charges is the **algebraic sum** $V = \sum kq_i/r_i$.

Potential difference (voltage): $\Delta V = V_B - V_A$. The work done **by an external agent** moving a charge q slowly from A to B is $W = q\Delta V$.

ƒ KEY RELATIONS

- Potential energy of two point charges: $U = k\dfrac{q_1 q_2}{r}$
- Uniform field (parallel plates): $\Delta V = Ed \Rightarrow E = \dfrac{V}{d}$
- **Electron-volt:** the energy gained by an electron accelerated through 1 V: $1\ \text{eV} = 1.6 \times 10^{-19}$ J.
- A charge accelerated from rest through ΔV: $\frac{1}{2}mv^2 = q\Delta V$.
- **Equipotential surfaces** are perpendicular to field lines; no work is done moving along them. The field points toward **decreasing** potential.

✎ WORKED EXAMPLE 1: ACCELERATING AN ELECTRON

An electron is accelerated from rest through 100 V. Find its kinetic energy and speed.

$K = e\Delta V = 100\ \text{eV} = 1.6 \times 10^{-17}$ J; $v = \sqrt{\dfrac{2K}{m}} = \sqrt{\dfrac{2(1.6 \times 10^{-17})}{9.11 \times 10^{-31}}} = 5.9 \times 10^{6}$ m/s.

✎ WORKED EXAMPLE 2: POTENTIAL FROM POTENTIAL ENERGY (AAU FINAL 2014)

Bringing $q = -8.8$ nC from infinity to a point 5.5 cm from a charge Q takes 13 mJ. Find the potential at that point.

$V = \dfrac{U}{q} = \dfrac{13 \times 10^{-3}}{-8.8 \times 10^{-9}} = -1.5 \times 10^{6}$ V $= -1.5$ MV. (So Q is negative.)

6.2.2 Capacitors

A **capacitor** is two conductors separated by an insulator; it **stores charge and energy**.

CAPACITANCE

Quantity	Formula
Definition	$C = \dfrac{Q}{V}$ (unit: farad, F = C/V; usually µF, nF, pF)
Parallel-plate capacitor	$C = \dfrac{\varepsilon_0 A}{d}$; with a dielectric: $C = \kappa\dfrac{\varepsilon_0 A}{d}$
Energy stored	$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \dfrac{Q^2}{2C}$
Parallel combination	$C_{eq} = C_1 + C_2 + \cdots$ (same V)
Series combination	$\dfrac{1}{C_{eq}} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \cdots$ (same Q)
Capacitors combine the **opposite way** to resistors.	

WORKED EXAMPLE 3: PARALLEL-PLATE CAPACITOR

Plates of area 0.020 m² are 1.0 mm apart in air and connected to 12 V. Find C, Q and U.

$C = \dfrac{(8.85\times10^{-12})(0.020)}{1.0\times10^{-3}} = 1.8\times10^{-10}$ F (177 pF); $Q = CV = 2.1\times10^{-9}$ C; $U = \frac{1}{2}CV^2 = 1.3\times10^{-8}$ J.

POTENTIAL TRAPS

- Potential is a **scalar**: add the values with signs, no vector components.
- Potential can be zero where the field is not (midway between $+q$ and $-q$), and vice versa.
- $E = V/d$ holds only for a **uniform** field.
- Inserting a dielectric **increases** C.

Exam Practice

7 QUESTIONS · LESSON 6.2

1 Wolaita Sodo University · Final Exam 2014

The potential difference between two points is 150 V. The work needed to move a 2 C charge between them is:

A) 200 J **B)** 150 J
C) 300 J **D)** 100 J

2 Addis Ababa University · Final 2014

A voltage V is applied between two points a distance d apart in a uniform electric field. The field is found by:

A) dividing the distance by the voltage
B) dividing the voltage by the square of the distance
C) multiplying the distance by the voltage
D) dividing the voltage by the distance

3 Addis Ababa University · Final 2014

A positive charge is at $x = 0$. A negative charge is then placed at $x = +3$ cm. What happens to the electric potential at $x = +1$ cm?

A) It decreases
B) It increases
C) It becomes zero
D) It does not change

4 Addis Ababa University · Final 2014

The energy required to bring a charge $q = -8.8$ nC from infinity to 5.5 cm from a point charge Q is 13 mJ. The potential at the final position of q is:

A) −1.12 MV
B) +1.5 MV
C) −0.66 MV
D) −1.5 MV

5 Model question

Two capacitors of 3 μF and 6 μF are connected in series. The equivalent capacitance is:

A) 9 μF
B) 18 μF
C) 4.5 μF
D) 2 μF

6 Model question

A proton is accelerated from rest through a potential difference of 5000 V. Find its final kinetic energy (in eV and J) and its speed.

7 Serway / OpenStax-style problem

A 10 μF capacitor is charged to 200 V. Find (a) the charge on each plate and (b) the energy stored.

Answers & Explanations

1. **C**

300 J

$W = q\Delta V = 2 \times 150 = 300$ J.

2. D

dividing the voltage by the distance

In a uniform field, $E = V/d$ (unit V/m).

3. A

It decreases

The negative charge adds a negative term kq/r to the (scalar) potential, so the total decreases.

4. D

−1.5 MV

$V = U/q = 0.013/(-8.8 \times 10^{-9}) = -1.48 \times 10^{6}$ V ≈ −1.5 MV.

5. D

2 μF

$\frac{1}{C} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2} \Rightarrow C = 2$ μF. (In parallel it would be 9 μF.)

6. **5000 eV = 8.0×10^{-16} J; 9.8×10^{5} m/s**

$K = e\Delta V = 5000$ eV $= 8.0 \times 10^{-16}$ J; $v = \sqrt{2K/m_p} = \sqrt{2(8.0 \times 10^{-16})/1.67 \times 10^{-27}} = 9.8 \times 10^{5}$ m/s.

7. **(a) 2.0×10^{-3} C; (b) 0.20 J**

$Q = CV = (10 \times 10^{-6})(200) = 2.0 \times 10^{-3}$ C; $U = \frac{1}{2}CV^2 = \frac{1}{2}(10 \times 10^{-6})(200)^2 = 0.20$ J.

CHAPTER 6 · LESSON 6.3

Current, Resistance, Ohm's Law and Electric Power

AFTER THIS LESSON YOU CAN

- Define electric current and drift velocity
- Use Ohm's law and resistivity
- Explain how resistance depends on length, area and temperature
- Calculate electric power and energy cost

6.3.1 Electric current

D CURRENT

The **rate of flow of charge** through a cross-section:

$$I = \frac{\Delta Q}{\Delta t} \qquad (\text{unit: ampere, } 1\ \text{A} = 1\ \text{C/s})$$

Conventional current flows from **+ to –** (the direction positive charges would move); in a metal the electrons actually drift the opposite way. Microscopically, $I = nqv_dA$, where n is the number of carriers per m³ and v_d the **drift velocity** (very slow, ≈ mm/s; the signal travels near the speed of light).

WORKED EXAMPLE 1: COUNTING ELECTRONS

A current of 3.0 A flows for 2.0 min. How much charge passes, and how many electrons?
$Q = It = 3.0 \times 120 = 360$ C; $n = Q/e = 360/1.6 \times 10^{-19} = 2.25 \times 10^{21}$ electrons.

6.3.2 Resistance and Ohm's law

D OHM'S LAW

For an **ohmic** conductor at constant temperature, the current is proportional to the voltage:

$$V = IR \qquad (R \text{ in ohms, } 1\ \Omega = 1\ \text{V/A})$$

Resistance measures how much a conductor opposes current. Diodes and bulbs are **non-ohmic** (their V–I graph is not a straight line).

RESISTIVITY

$$R = \rho \frac{L}{A}$$

- ρ = **resistivity** (Ω·m), a property of the **material**: copper 1.7×10^{-8}, aluminium 2.8×10^{-8}, nichrome 1.0×10^{-6} Ω·m. Conductivity $\sigma = 1/\rho$.
- Longer wire → **more** resistance; thicker wire → **less** resistance ($A = \pi d^2/4$).
- For metals R **increases** with temperature: $R = R_0[1 + \alpha(T - T_0)]$. For semiconductors it **decreases**.

WORKED EXAMPLE 2: A COPPER WIRE

Find the resistance of 10 m of copper wire of diameter 1.0 mm.

$A = \pi(0.50 \times 10^{-3})^2 = 7.85 \times 10^{-7}$ m² → $R = \dfrac{(1.7 \times 10^{-8})(10)}{7.85 \times 10^{-7}} = 0.22\ \Omega$.

6.3.3 Electric power and energy

$$P = IV = I^2R = \frac{V^2}{R} \qquad \text{(watts)}$$

Energy used: $E = Pt$. The electricity company bills in **kilowatt-hours**: 1 kWh = 3.6×10^6 J. A cell of emf ε that drives a charge Q round a circuit supplies **total energy** εQ.

WORKED EXAMPLE 3: A LIGHT BULB

A bulb is rated 60 W, 220 V. Find its current, resistance, and the energy it uses in 5.0 h.

$I = P/V = 0.27$ A; $R = V^2/P = 807\ \Omega$; $E = (0.060\ \text{kW})(5.0\ \text{h}) = 0.30$ kWh $= 1.1 \times 10^6$ J.

CURRENT AND POWER TRAPS

- $P = V^2R$ is **wrong** (dimensionally). The three correct forms are IV, I^2R and V^2/R.
- Doubling both L and d of a wire **halves** R: L × 2, but A × 4.
- Same voltage, twice the current → **half** the resistance.
- kWh is a unit of **energy**, not power.

Exam Practice

7 QUESTIONS · LESSON 6.3

1 Addis Ababa University · Final 2014

Which equation for electric power is INCORRECT?

A) $P = I^2R$
B) $P = V^2/R$
C) $P = IV$
D) $P = V^2R$

2 Addis Ababa University · Final 2014

A 9.0 V battery is connected across resistor R_1, and then across R_2; with R_2 the current is twice as large. The ratio R_1/R_2 is:

A) 2 **B)** 0.5
C) 4 **D)** 1

3 Addis Ababa University · Final 2014

A cylindrical rod has resistance R. If both its length and its diameter are doubled, its new resistance is:

A) $R/2$ **B)** R
C) $2R$ **D)** $4R$

4 Addis Ababa University · Final 2014

A cell of emf ε delivers a charge Q to an external circuit. Which statement is correct?

A) The energy dissipated in the external circuit is εQ
B) The energy dissipated within the cell is εQ
C) The total energy delivered by the cell is εQ
D) The external resistance is εQ

5 Model question

For metals, as the temperature increases, the resistance:

A) increases **B)** decreases
C) stays the same **D)** becomes zero

6 Model question

A 1500 W electric kettle is used for 20 minutes each day for 30 days. How much energy does it use, and what does it cost at 2.0 birr per kWh?

7 Serway / OpenStax-style problem

A nichrome wire ($\rho = 1.0 \times 10^{-6}$ Ω·m) of radius 0.32 mm is used in a heater. What length gives a resistance of 10 Ω? What power does it deliver on 120 V?

Answers & Explanations

1. **D**

$P = V^2 R$

Substituting $I = V/R$ into $P = IV$ gives V^2/R, not $V^2 R$.

2. **A**

2

$R = V/I$ with the same V: twice the current means half the resistance, so $R_1 = 2R_2$.

3. **A**

$R/2$

$R = \rho L/A$, $A \propto d^2$: $\dfrac{2L}{4A} = \tfrac{1}{2}\dfrac{L}{A}$.

4. **C**

The total energy delivered by the cell is εQ

The emf is the energy supplied per unit charge, so the total energy is εQ, shared between the internal resistance and the external circuit.

5. **A**

increases

Hotter atoms vibrate more and scatter the electrons more, so R rises. (Semiconductors do the opposite.)

6. **15 kWh; 30 birr**

$E = (1.5\ \text{kW})(\tfrac{1}{3}\ \text{h})(30) = 15$ kWh; cost $= 15 \times 2.0 = 30$ birr.

7. **3.2 m; 1.44 kW**

$A = \pi(0.32 \times 10^{-3})^2 = 3.2 \times 10^{-7}$ m²; $L = RA/\rho = 10(3.2 \times 10^{-7})/(1.0 \times 10^{-6}) = 3.2$ m.

$P = V^2/R = 14\,400/10 = 1440$ W.

CHAPTER 6 · LESSON 6.4

Resistor Combinations, EMF and Kirchhoff's Rules

AFTER THIS LESSON YOU CAN

- Find the equivalent resistance of series and parallel combinations
- Use emf and internal resistance
- State Kirchhoff's junction and loop rules
- Solve multi-loop circuits

6.4.1 Series and parallel resistors

COMBINATIONS

	Series	Parallel
Same for all	**current** I	**voltage** V
Divides	voltage: $V = V_1 + V_2 + \cdots$	current: $I = I_1 + I_2 + \cdots$
Equivalent	$R_{eq} = R_1 + R_2 + \cdots$ (larger than any one)	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$ (smaller than any one)
Two resistors		$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$
Example	old Christmas lights (one fails, all go out)	**household wiring** (each appliance gets full voltage, works independently)
n equal resistors R: series nR; parallel R/n.		

WORKED EXAMPLE 1: MIXED CIRCUIT

$R_1 = 2\ \Omega$ is in series with a parallel pair $R_2 = 3\ \Omega$ and $R_3 = 6\ \Omega$, across a 12 V battery. Find R_{eq} and every current and voltage.

Parallel pair: $\frac{3 \times 6}{3 + 6} = 2\ \Omega$ → $R_{eq} = 2 + 2 = 4\ \Omega$.

Total current $I = 12/4 = 3$ A (through R_1). $V_1 = 3 \times 2 = 6$ V; the parallel pair has $12 - 6 = 6$ V.

$I_2 = 6/3 = 2$ A; $I_3 = 6/6 = 1$ A (check: $2 + 1 = 3$ A ✓).

6.4.2 EMF and internal resistance

The **emf** ε of a battery is the energy it gives each coulomb (its voltage when no current flows). Real batteries have **internal resistance** r, so the **terminal voltage** drops when current flows:

$$I = \frac{\varepsilon}{R + r}, \qquad V_{\text{terminal}} = \varepsilon - Ir = IR$$

WORKED EXAMPLE 2: INTERNAL RESISTANCE (AAU FINAL 2014)

A 20 V battery with $r = 1.5\ \Omega$ drives a 10 Ω resistor. Find the voltage across the resistor.
$I = \dfrac{20}{10 + 1.5} = 1.74$ A → $V = IR = 17.4$ V (the other 2.6 V is lost inside the battery).

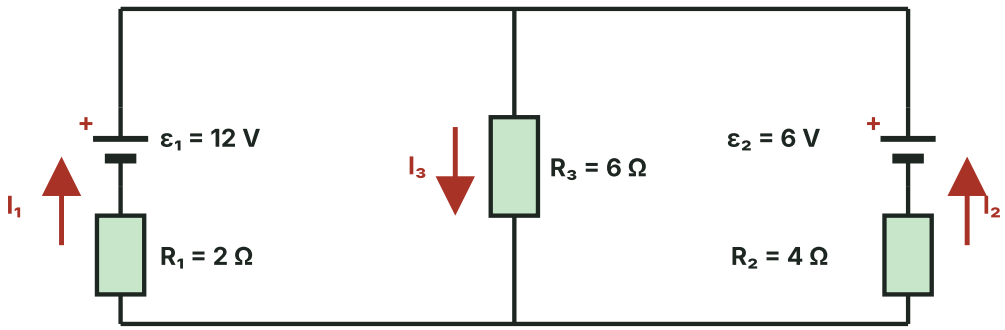
6.4.3 Kirchhoff's rules

D KIRCHHOFF'S TWO RULES

1. **Junction rule** (conservation of **charge**): the total current **into** a junction equals the total current **out**: $\sum I_{\text{in}} = \sum I_{\text{out}}$.
2. **Loop rule** (conservation of **energy**): around any closed loop, the sum of the potential changes is **zero**: $\sum \Delta V = 0$.

★ SIGN RULES FOR THE LOOP RULE

- Crossing a **resistor in the direction of the current**: $-IR$ (potential drops). Against the current: $+IR$.
- Crossing a **battery from – to +**: $+\varepsilon$. From + to –: $-\varepsilon$.
- Guess current directions freely; a **negative answer** just means the current flows the other way.



Two-loop circuit: both batteries push current up their branches; I_3 flows down the middle

✎ WORKED EXAMPLE 3: A TWO-LOOP CIRCUIT

Solve the circuit above for I_1, I_2, I_3.
Junction (top): $I_3 = I_1 + I_2$.
Left loop (up through ε_1, down through R_3, clockwise): $12 - 2I_1 - 6I_3 = 0$.
Right loop (up through ε_2, down through R_3): $6 - 4I_2 - 6I_3 = 0$.
From these: $I_1 = 6 - 3I_3$ and $I_2 = 1.5 - 1.5I_3$. Then $I_3 = 7.5 - 4.5I_3 \Rightarrow I_3 = 1.36$ A.
$I_1 = 1.91$ A; $I_2 = -0.55$ A (**negative**: it actually flows **down** through the 6 V battery, which is being charged).
Check the right loop: $6 - 4(-0.55) - 6(1.36) = 6 + 2.2 - 8.2 = 0$ ✓.

CIRCUIT TRAPS

- Parallel R_{eq} is **smaller than the smallest** resistor.
- An ideal ammeter goes in **series** (≈ 0 Ω); an ideal voltmeter in **parallel** (≈ ∞ Ω).
- The terminal voltage of a real battery is less than its emf when it delivers current.
- Junction rule = charge conservation; loop rule = energy conservation.

Exam Practice

6 QUESTIONS · LESSON 6.4

1 Addis Ababa University · Final 2014

A battery of emf 20 V and internal resistance 1.5 Ω is connected in series with a 10 Ω resistor. The potential difference across the resistor is:

A) 17.4 V **B)** 20 V
C) 2.6 V **D)** 11.5 V

2 Model question

Three 6 Ω resistors are connected in parallel. The equivalent resistance is:

A) 18 Ω **B)** 3 Ω
C) 6 Ω **D)** 2 Ω

3 Model question

Kirchhoff's junction rule is a statement of the conservation of:

A) charge **B)** momentum
C) energy **D)** mass

4 Model question

Household appliances are connected in parallel because:

A) each appliance gets the full mains voltage and can be switched independently
B) it uses less current
C) the total resistance increases
D) it is cheaper

5 Model question (exam format)

$R_1 = 2\ \Omega$ is in series with a parallel combination of $R_2 = 3\ \Omega$ and $R_3 = 6\ \Omega$, connected to a 12 V battery. Find (i) R_{eq}, (ii) the currents through R_2 and R_3, (iii) the voltage across R_1.

6 Serway / OpenStax-style problem

In the two-loop circuit of lesson 6.4 (ε_1 = 12 V, ε_2 = 6 V, R_1 = 2 Ω, R_2 = 4 Ω, R_3 = 6 Ω), what power is dissipated in R_3?

Answers & Explanations

1. **A**

17.4 V

$I = 20/11.5 = 1.74$ A; $V = IR = 17.4$ V.

2. **D**

2 Ω

n equal resistors in parallel: $R/n = 6/3 = 2$ Ω.

3. **A**

charge

Charge cannot pile up at a junction, so current in = current out. The loop rule expresses conservation of energy.

4. **A**

each appliance gets the full mains voltage and can be switched independently

In parallel every branch has the same voltage and works independently of the others.

5. **(i) 4 Ω; (ii) 2 A and 1 A; (iii) 6 V**

Parallel pair = 2 Ω, so $R_{eq} = 4$ Ω and $I = 3$ A. $V_1 = 6$ V; the pair has 6 V, giving 2 A and 1 A.

6. **11 W**

$I_3 = 1.36$ A → $P = I_3^2 R_3 = (1.36)^2(6) = 11.2$ W.

CHAPTER 6 · LESSON 6.5

Magnetic Field, Magnetic Force and Magnetic Flux

AFTER THIS LESSON YOU CAN

- Describe magnets, magnetic poles and field lines
- Calculate the magnetic force on a moving charge and on a current-carrying wire
- Find the field of a long straight wire and a solenoid
- Define and calculate magnetic flux

6.5.1 Magnets and magnetic fields

D BASIC FACTS

- Every magnet has a **north** and a **south pole**. **Like poles repel; unlike poles attract.**
- Magnetic poles **always come in pairs**: cutting a magnet in half gives two smaller magnets (no isolated "monopole" has been found).
- **Moving charges (currents) are the source of all magnetism**, from electromagnets to the spinning electrons in iron.
- The magnetic field $\vec{B}$ is measured in **tesla** (T). 1 gauss = 10^{-4} T; the Earth's field is about 5×10^{-5} T.
- Field lines leave the **N pole** and enter the **S pole** outside the magnet; they form **closed loops** and never cross.

Applications: compasses, electric motors and generators, loudspeakers, hard drives, MRI scanners, maglev trains, electromagnets that lift scrap iron.

6.5.2 Force on a moving charge

F MAGNETIC (LORENTZ) FORCE

$$\vec{F} = q\vec{v} \times \vec{B}, \qquad F = |q| v B \sin\theta$$

- The force is **perpendicular to both** $\vec{v}$ and $\vec{B}$ (right-hand rule; reverse it for a negative charge).
- **No force** if the charge is **at rest** ($v = 0$) or moves **parallel** to $\vec{B}$ ($\theta = 0^\circ$ or 180°). Maximum when $\vec{v} \perp \vec{B}$.
- Because $\vec{F} \perp \vec{v}$, the magnetic force does **no work**: it changes the **direction**, not the speed.
- A charge moving perpendicular to a uniform field moves in a **circle**: $qvB = \dfrac{mv^2}{r} \Rightarrow r = \dfrac{mv}{qB}$.

✎ WORKED EXAMPLE 1: A PROTON IN A MAGNETIC FIELD

A proton moves at 2.0×10^6 m/s perpendicular to a 0.50 T field. Find the force and the radius of its path.

$F = qvB = (1.6 \times 10^{-19})(2.0 \times 10^6)(0.50) = 1.6 \times 10^{-13}$ N.

$r = \dfrac{mv}{qB} = \dfrac{(1.67 \times 10^{-27})(2.0 \times 10^6)}{(1.6 \times 10^{-19})(0.50)} = 4.2 \times 10^{-2}$ m = 4.2 cm.

6.5.3 Force on a current-carrying wire

$$F = BIL\sin\theta$$

where θ is the angle between the wire (current direction) and $\vec{B}$. This force drives **electric motors** and **loudspeakers**.

WORKED EXAMPLE 2: WIRE IN A FIELD

A 0.50 m wire carries 4.0 A at 30° to a 0.20 T field. $F = (0.20)(4.0)(0.50)\sin 30^\circ = 0.20$ N.

6.5.4 Magnetic fields produced by currents

F FIELDS OF CURRENTS

Source	Field	Direction
Long straight wire	$B = \dfrac{\mu_0 I}{2\pi r}$	circles around the wire (right-hand grip rule)
Centre of a circular loop (N turns, radius R)	$B = \dfrac{\mu_0 N I}{2R}$	along the axis
Long solenoid (n turns per metre)	$B = \mu_0 n I$ (uniform inside)	along the axis

$\mu_0 = 4\pi \times 10^{-7}$ T·m/A. Two parallel wires with currents in the **same** direction **attract**; opposite directions **repel**.

WORKED EXAMPLE 3: FIELD OF A WIRE

Find B at 5.0 cm from a long wire carrying 10 A. $B = \dfrac{(4\pi \times 10^{-7})(10)}{2\pi(0.050)} = 4.0 \times 10^{-5}$ T (comparable to the Earth's field).

6.5.5 Magnetic flux

D MAGNETIC FLUX

The amount of magnetic field passing **through** a surface:

$$\Phi_B = BA\cos\theta \qquad (\text{unit: weber, } 1\ \text{Wb} = 1\ \text{T}\cdot\text{m}^2)$$

where θ is the angle between $\vec{B}$ and the **normal** (perpendicular) to the surface. Maximum when $\vec{B}$ is perpendicular to the surface ($\theta = 0$); zero when $\vec{B}$ is parallel to the surface ($\theta = 90^\circ$).

WORKED EXAMPLE 4: FLUX THROUGH A LOOP

A loop of area 0.050 m² is in a 0.40 T field whose direction makes 60° with the loop's normal.
$\Phi = (0.40)(0.050)\cos 60^\circ = 1.0 \times 10^{-2}$ Wb.

MAGNETISM TRAPS

- A charge **at rest** feels **no** magnetic force; neither does a charge moving **along** the field.
- The magnetic force never changes a particle's **speed** or kinetic energy.
- In $\Phi = BA\cos\theta$, θ is measured from the **normal**, not from the plane.
- Electric field lines start and stop on charges; magnetic field lines are **closed loops**.

Exam Practice

7 QUESTIONS · LESSON 6.5

1 Addis Ababa University · Final 2014

An electron is at rest in a magnetic field B, and a proton moves with speed v parallel to the field. The magnetic forces on them are:

A) proton: evB; electron: zero
B) proton: zero; electron: zero
C) proton: maximum; electron: zero
D) proton: zero; electron: evB

2 Wolaita Sodo University · Final Exam 2014

A magnetic field is a phenomenon that describes the magnetic force influence on moving electric charges inside the field.

True / False

3 Model question

The magnetic force on a moving charged particle:

A) increases its speed
B) is parallel to its velocity
C) does positive work on it
D) changes only the direction of its velocity

4 Model question

Two long parallel wires carry currents in the same direction. The wires:

A) attract each other
B) repel each other
C) exert no force on each other
D) rotate

5 Model question

The SI unit of magnetic flux is the:

A) weber
B) tesla
C) henry
D) gauss

6 Serway / OpenStax-style problem

An electron moving at 3.0×10^6 m/s enters a 0.020 T field at right angles. Find the force on it and the radius of its circular path.

7 Model question

A solenoid 0.40 m long has 800 turns and carries 2.0 A. Find the magnetic field inside it.

Answers & Explanations

1. **B**

proton: zero; electron: zero

$F = qvB\sin\theta$: the electron has $v = 0$, and the proton has $\theta = 0^\circ$, so both forces are zero.

2. **True**

A magnetic field exerts a force $q\vec{v} \times \vec{B}$ on moving charges (and on currents).

3. **D**

changes only the direction of its velocity

The force is always perpendicular to the velocity, so it does no work and only bends the path.

4. **A**

attract each other

Each wire sits in the other's magnetic field; for parallel currents, $F = BIL$ points toward the other wire.

5. **A**

weber

$\Phi = BA$: T·m² = Wb. The tesla is the unit of the field B.

6. **9.6 × 10⁻¹⁵ N; 8.5 × 10⁻⁴ m**

$F = evB = (1.6 \times 10^{-19})(3.0 \times 10^6)(0.020) = 9.6 \times 10^{-15}$ N.

$r = \dfrac{mv}{eB} = \dfrac{(9.11 \times 10^{-31})(3.0 \times 10^6)}{(1.6 \times 10^{-19})(0.020)} = 8.5 \times 10^{-4}$ m.

7. **5.0 × 10⁻³ T**

$n = 800/0.40 = 2000$ turns/m; $B = \mu_0 nI = (4\pi \times 10^{-7})(2000)(2.0) = 5.0 \times 10^{-3}$ T.

CHAPTER 6 · LESSON 6.6

Electromagnetic Induction

AFTER THIS LESSON YOU CAN

- State Faraday's law and Lenz's law
- Calculate induced emf from a changing flux and from a moving rod
- Explain generators and transformers
- Solve transformer problems

6.6.1 Faraday's law

D FARADAY'S LAW OF INDUCTION

An emf is induced in a circuit whenever the magnetic flux through it changes. The induced emf equals the rate of change of flux:

$$\varepsilon = -N\frac{\Delta\Phi_B}{\Delta t}$$

(N = number of turns). The flux can change because **B changes**, the **area changes**, or the **angle changes** (rotation).

E HOW TO INDUCE A CURRENT WITH A COIL AND A MAGNET

Move the **magnet** into or out of the coil, **or** move the **coil**; it is the **relative motion** (change of flux) that matters. Holding them still, or moving them **together**, induces nothing. A stronger magnet, faster motion and more turns give a larger emf.

D LENZ'S LAW

The induced current flows in the direction that opposes the change in flux that produced it. (This is the meaning of the minus sign in Faraday's law, and it follows from conservation of energy.) Push a N pole into a coil: the coil's near end becomes a **N pole** to repel it.

WORKED EXAMPLE 1: A CHANGING FIELD

A 200-turn coil of area 0.010 m² is perpendicular to a field that rises from 0 to 0.50 T in 0.10 s. Find the average induced emf.

$\varepsilon = N\dfrac{\Delta(BA)}{\Delta t} = 200 \times \dfrac{(0.50)(0.010)}{0.10} = 10$ V.

6.6.2 Motional emf

A rod of length L moving with speed v perpendicular to a field B has an emf between its ends:

$$\varepsilon = BLv$$

Inside the rod each electron feels $F = evB$, which pushes electrons to one end until the electric field $E = vB$ balances it.

WORKED EXAMPLE 2: MOVING ROD (AAU FINAL 2014)

A 25 cm rod moves at 5.0 m/s perpendicular to a 0.25 T field. Find (a) the magnetic force on an electron in the rod, (b) the electric field in the rod, (c) the potential difference between its ends, (d) the speed needed for 1.0 V.

(a) $F = evB = (1.6 \times 10^{-19})(5.0)(0.25) = 2.0 \times 10^{-19}$ N.

(b) Balance: $eE = evB \Rightarrow E = vB = 1.25$ V/m.

(c) $\varepsilon = BLv = (0.25)(0.25)(5.0) = 0.31$ V.

(d) $v = \dfrac{\varepsilon}{BL} = \dfrac{1.0}{(0.25)(0.25)} = 16$ m/s.

6.6.3 Generators and transformers

AC GENERATOR

A coil rotating in a magnetic field has a continuously changing flux, so it produces an **alternating emf**: $\varepsilon = NBA\omega \sin \omega t$, with peak value $NBA\omega$. It converts **mechanical energy to electrical energy** (hydroelectric dams, wind turbines). A **motor** does the reverse.

TRANSFORMER

Two coils on an iron core. An **alternating** current in the primary makes a changing flux, which induces an emf in the secondary:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}, \qquad I_p V_p = I_s V_s \ \text{(ideal)}$$

Step-up ($N_s > N_p$): voltage up, current down. **Step-down**: the reverse. Transformers **do not work on DC** (no changing flux).

WORKED EXAMPLE 3: PHONE CHARGER TRANSFORMER

A transformer steps 220 V down to 11 V. The primary has 1000 turns. How many turns are on the secondary? If the secondary delivers 2.0 A, what is the primary current (ideal)?

$N_s = 1000 \times \dfrac{11}{220} = 50$ turns; $I_p = \dfrac{I_s V_s}{V_p} = \dfrac{(2.0)(11)}{220} = 0.10$ A.

Power is transmitted over long distances at **high voltage** and low current because the heating loss I^2R in the lines is then small.

INDUCTION TRAPS

- A **steady** field through a stationary coil induces **nothing**; the flux must **change**.
- Transformers need **AC**.
- A step-up transformer raises voltage but **lowers current**; it cannot create energy.
- Lenz's law: the induced current **opposes the change**, not the field itself.

Exam Practice

6 QUESTIONS · LESSON 6.6

1 Addis Ababa University · Final 2014

To produce a current with a coil and a bar magnet you can:

A) put the coil close to the magnet
B) move both the coil and the magnet together
C) move either the coil or the magnet
D) A and B
E) B and C

2 Model question

The direction of an induced current is always such that it opposes the change producing it. This is:

A) Faraday's law
B) Ampère's law
C) Ohm's law
D) Lenz's law

3 Model question

A transformer has 100 turns on the primary and 2000 turns on the secondary. If 12 V AC is applied to the primary, the secondary voltage is:

A) 0.6 V
B) 12 V
C) 2400 V
D) 240 V

4 Model question

A transformer connected to a 12 V car battery (DC) gives an output of:

A) zero (after the instant of switching)
B) a smaller DC voltage
C) a larger DC voltage
D) AC

5 Addis Ababa University · Final 2014

A 25 cm rod moves at 5.0 m/s in a plane perpendicular to a 0.25 T magnetic field directed into the page. Calculate (a) the magnetic force on an electron in the rod, (b) the electric field in the rod, (c) the potential difference between the ends, (d) the speed needed for a potential difference of 1.0 V.

6 Serway / OpenStax-style problem

A 50-turn circular coil of radius 5.0 cm lies perpendicular to a magnetic field that decreases uniformly from 0.60 T to 0.20 T in 0.20 s. Find the induced emf.

Answers & Explanations

1. **C**

move either the coil or the magnet

An emf needs a **changing flux**, i.e. relative motion. Holding them still or moving them together keeps the flux constant.

2. **D**

Lenz's law

Lenz's law gives the direction (the minus sign in Faraday's law); Faraday's law gives the size.

3. **D**

240 V

$V_s = V_p \dfrac{N_s}{N_p} = 12 \times 20 = 240$ V (step-up).

4. **A**

zero (after the instant of switching)

DC gives a constant flux, so no emf is induced once the current is steady.

5. **(a) 2.0×10^{-19} N; (b) 1.25 V/m; (c) 0.31 V; (d) 16 m/s**

(a) evB; (b) $E = vB$; (c) $\varepsilon = BLv = 0.3125$ V; (d) $v = \varepsilon/(BL)$.

6. **0.79 V**

$A = \pi(0.050)^2 = 7.85 \times 10^{-3}$ m²; $\varepsilon = N\dfrac{A\Delta B}{\Delta t} = 50 \times \dfrac{(7.85 \times 10^{-3})(0.40)}{0.20} = 0.79$ V.

CHAPTER 6 · LESSON 6.7

Conductors, Insulators and Semiconductors

AFTER THIS LESSON YOU CAN

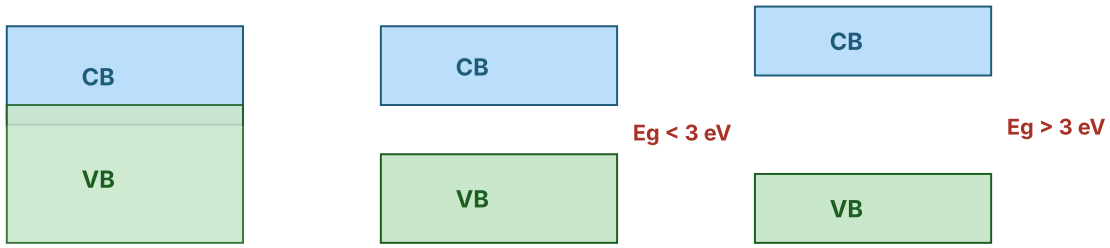
- Explain energy bands and the band gap
- Classify materials as conductors, insulators and semiconductors
- Distinguish intrinsic and extrinsic semiconductors
- Explain n-type and p-type doping

6.7.1 Energy bands in solids

When atoms join to form a solid, their outer energy levels overlap and spread into **bands** of closely spaced levels.

D THE THREE BANDS

- **Valence band (VB):** occupied by the valence (outermost) electrons; it can be partly or completely filled, never empty.
- **Conduction band (CB):** the next allowed band above it. Electrons here are **free** and carry current. It may be empty or partly filled, never completely filled.
- **Forbidden energy gap (E_g):** the energy range between the top of the VB and the bottom of the CB, where no electron states exist. **Its size decides whether a material conducts.**



Band pictures of a conductor, a semiconductor and an insulator

6.7.2 Classifying materials

Property	Conductor	Semiconductor	Insulator
Band gap	**none** (bands overlap)	**small**, < 3 eV (Si 1.1 eV, Ge 0.7 eV)	**large**, > 3 eV (diamond ≈ 6 eV)
Conductivity ($\Omega^{-1}m^{-1}$)	10^4 to 10^7	10^{-6} to 10^4	10^{-20} to 10^{-10}
Free electrons at room temperature	very many	a few	almost none

Property	Conductor	Semiconductor	Insulator
Effect of **raising temperature**	conductivity **decreases**	conductivity **increases**	almost no effect
Effect of **impurities**	little effect	**large increase** (doping)	little effect
Examples	Cu, Al, Ag, Au	Si, Ge, GaAs	glass, wood, plastic, mica

6.7.3 Intrinsic semiconductors

A **pure** semiconductor crystal (e.g. silicon, with each atom sharing 4 covalent bonds) is **intrinsic**. At room temperature a few bonds break: an electron jumps to the CB and leaves behind a **hole** in the VB. A **hole** behaves as a **positive charge carrier**. In an intrinsic semiconductor the number of free electrons **equals** the number of holes. At absolute zero it is an insulator.

6.7.4 Extrinsic semiconductors: doping

Doping is adding a tiny amount of impurity (about 1 atom per 10^6) to a pure semiconductor to greatly increase its conductivity.

★ N-TYPE

- Doped with a **pentavalent** (group V, 5 valence electrons) impurity: phosphorus (P), arsenic (As), antimony (Sb).
- The fifth electron is free: the impurity is a **donor**.
- **Majority carriers: electrons**; minority carriers: holes.

D P-TYPE

- Doped with a **trivalent** (group III, 3 valence electrons) impurity: boron (B), aluminium (Al), gallium (Ga), indium (In).
- One bond is incomplete, leaving a **hole**: the impurity is an **acceptor**.
- **Majority carriers: holes**; minority carriers: electrons.

! SEMICONDUCTOR TRAPS

- n-type and p-type materials are both **electrically neutral** overall. "n" and "p" name the **majority carrier**, not the net charge.
- Heating a **metal** lowers its conductivity; heating a **semiconductor** raises it.
- Pentavalent → donor → n-type. Trivalent → acceptor → p-type. ("**P**enta gives **N**; **T**ri gives **P**", or remember that **D**onors make **N**egative carriers.)

Exam Practice

6 QUESTIONS · LESSON 6.7

1 Model question (module)

Materials are classified as conductors, semiconductors and insulators according to:

A) their density
B) the size of the forbidden energy gap
C) their colour
D) their melting point

2 Model question (module)

When the temperature of a semiconductor increases, its conductivity:

A) increases
B) decreases
C) stays the same
D) drops to zero

3 Model question (module)

Adding arsenic (a pentavalent atom) to pure silicon produces:

A) a p-type semiconductor with holes as majority carriers
B) an intrinsic semiconductor
C) an insulator
D) an n-type semiconductor with electrons as majority carriers

4 Model question (module)

In a p-type semiconductor, the majority charge carriers are:

A) electrons
B) holes
C) protons
D) positive ions

5 Model question (module)

An n-type semiconductor carries a net negative charge.

True / False

6 Model question (module)

The forbidden energy gap of silicon is about:

A) 0 eV
B) 0.1 eV
C) 1.1 eV
D) 6 eV

Answers & Explanations

1. B

the size of the forbidden energy gap

Overlapping bands → conductor; small gap (< 3 eV) → semiconductor; large gap (> 3 eV) → insulator.

2. A

increases

More electrons gain enough thermal energy to cross the small gap, creating more electron–hole pairs.

3. D

an n-type semiconductor with electrons as majority carriers

Arsenic's fifth electron is donated to the crystal, so electrons become the majority carriers.

4. B

holes

A trivalent acceptor leaves an incomplete bond, i.e. a hole, which acts as a positive carrier.

5. False

The donor atoms are neutral when added; the crystal as a whole stays neutral. "n" refers only to the majority carriers (electrons).

6. C

1.1 eV

Si: 1.1 eV; Ge: about 0.7 eV; diamond (an insulator): about 6 eV.

CHAPTER 6 · LESSON 6.8

Diodes and Transistors

AFTER THIS LESSON YOU CAN

- Explain the p–n junction, depletion region and barrier potential
- Describe forward and reverse bias and the diode I–V curve
- Explain half-wave and full-wave rectification
- Describe the structure, biasing, configurations and uses of transistors

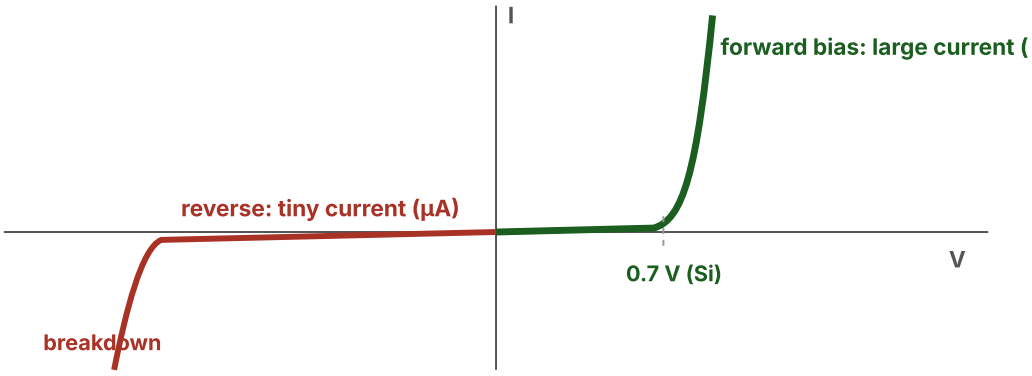
6.8.1 The p–n junction diode

When p-type and n-type regions are formed in one crystal, electrons from the n side **diffuse** across and fill holes on the p side (and holes diffuse the other way). Near the junction this leaves a thin layer (10^{-6} to 10^{-4} cm) with **no mobile carriers**, only fixed ions: the **depletion region**. Its fixed charges (– on the p side, + on the n side) create the **barrier potential**: about **0.7 V for silicon** and **0.3 V for germanium**.

BIASING A DIODE

	Forward bias	Reverse bias
Connection	**p → +** terminal, **n → –** terminal	p → –, n → +
Depletion layer	becomes **narrower**	becomes **wider**
Barrier	reduced; overcome above ~0.7 V (Si)	increased
Current	**large** (mA), rises steeply	tiny leakage (μA) from minority carriers
Diode acts as	a closed switch (**ON**)	an open switch (**OFF**)

If the reverse voltage is made very large, the junction **breaks down** and conducts heavily (used deliberately in **Zener diodes** for voltage regulation).



Current–voltage characteristic of a silicon diode (non-ohmic)

The diode symbol is an arrow pointing in the direction of **conventional current** (p → n) with a bar at the n side (cathode). A diode lets current through in **one direction only**, so it protects devices if batteries are inserted backwards.

6.8.2 Rectification

Rectification converts **AC** (alternating) into **DC** (one-direction) current.

★ HALF-WAVE RECTIFIER

One diode. It conducts during the **positive** half-cycles and blocks the negative half-cycles, so the output is a series of positive humps with gaps. Simple, but wasteful.

E FULL-WAVE (BRIDGE) RECTIFIER

Four diodes in a bridge. During **both** half-cycles current flows through the load **in the same direction** (through two diodes at a time), so both halves are used. A **capacitor** across the load then smooths the output into steady DC (as in phone chargers).

Other diodes: **LED** (light-emitting diode: gives light when forward biased), **photodiode** and **solar cell** (light produces current), **Zener diode** (voltage regulator).

6.8.3 Transistors

A diode cannot **amplify**. A **transistor** can. A **bipolar junction transistor (BJT)** is a sandwich of three doped layers forming two p–n junctions back to back: **npn** or **pnp**.

D THE THREE REGIONS

Region	Doping	Size	Job
Emitter (E)	**heavily** doped	medium	supplies (emits) the majority carriers
Base (B)	**lightly** doped	**very thin**	controls the flow
Collector (C)	moderately doped	**largest**	collects the carriers (dissipates heat)

In the symbol, the **arrow is on the emitter** and points in the direction of conventional current (out for npn, in for pnp: "**N**ot **P**ointing i**N**" = npn).

★ BIASING FOR AMPLIFICATION

For normal (**active**) operation: the **emitter–base junction is forward biased** and the **collector–base junction is reverse biased**.

$$I_E = I_B + I_C, \qquad \alpha = \frac{I_C}{I_E}\ (\text{slightly less than } 1), \qquad \beta = \frac{I_C}{I_B}\ (\text{typically } 20 \text{ to } 300)$$

A small base current controls a **large** collector current: this is **current amplification**. $\beta = \dfrac{\alpha}{1-\alpha}$.

Configurations (named after the lead common to input and output): **common base (CB)**, **common emitter (CE)** and **common collector (CC)**. The **CE amplifier** is the most widely used: it gives high current and voltage gain, and its output voltage is **180° out of phase** with the input.

✎ WORKED EXAMPLE 1: TRANSISTOR CURRENTS (MODULE)

In a common-base connection, $I_E = 1.0$ mA and $I_C = 0.95$ mA. Find I_B, α and β.

$I_B = I_E - I_C = 0.05$ mA $= 50$ μA; $\alpha = 0.95/1.0 = 0.95$; $\beta = I_C/I_B = 0.95/0.05 = 19$.

✎ WORKED EXAMPLE 2: CURRENT GAIN

A transistor with β = 100 has a base current of 40 μA. Find I_C and I_E.

$I_C = \beta I_B = 100 \times 40$ μA $= 4.0$ mA; $I_E = I_B + I_C = 4.04$ mA.

E USES OF TRANSISTORS

Amplifiers (radios, microphones, hearing aids: a weak signal controls a larger one) and **switches** (a small base current turns a large collector current fully ON or OFF). Billions of tiny transistor switches on one chip form the **logic gates** and memory of every computer and phone.

DIODE AND TRANSISTOR TRAPS

- Forward bias: **p to positive**. The depletion layer **shrinks** in forward bias and **widens** in reverse bias.
- The barrier potential (0.7 V Si, 0.3 V Ge) is not a voltage you can measure with a voltmeter across an unconnected diode.
- In a transistor the **base** is the thinnest and most lightly doped; the **emitter** is the most heavily doped.
- I_E is the largest current: $I_E = I_B + I_C$.

Exam Practice

8 QUESTIONS · LESSON 6.8

1 Model question (module)

A p–n junction diode is forward biased when:

A) the p side is connected to the negative terminal
B) the p side is connected to the positive terminal and the n side to the negative terminal
C) both sides are connected to the positive terminal
D) no battery is connected

2 Model question (module)

In reverse bias, the depletion region of a p–n junction:

A) disappears
B) becomes narrower
C) becomes wider
D) stays the same

3 Model question (module)

The barrier potential of a silicon diode is about:

A) 0.3 V
B) 1.1 V
C) 0.7 V
D) 3 V

4 Model question (module)

The process of converting AC into DC is called:

A) amplification
B) rectification
C) modulation
D) induction

5 Model question (module)

In a bipolar transistor, the region that is most heavily doped is the:

A) base
B) collector
C) emitter
D) all equally

6 Model question (module)

For a transistor to work as an amplifier:

A) both junctions must be forward biased
B) both junctions must be reverse biased
C) the emitter–base junction is forward biased and the collector–base junction is reverse biased
D) the emitter–base junction is reverse biased and the collector–base junction is forward biased

7 Model question (module example)

In a common-base transistor circuit, $I_E = 1.0$ mA and $I_C = 0.95$ mA. Find the base current and the current gains α and β.

8 Serway / OpenStax-style problem

A transistor has β = 150. What base current is needed for a collector current of 30 mA, and what is the emitter current?

Answers & Explanations

1. B

the p side is connected to the positive terminal and the n side to the negative terminal

Forward bias: p → +, n → −. This narrows the depletion layer and lets a large current flow.

2. C

becomes wider

The battery pulls holes and electrons away from the junction, widening the region with no mobile carriers.

3. C

0.7 V

Si ≈ 0.7 V; Ge ≈ 0.3 V. (1.1 eV is silicon's band gap, a different quantity.)

4. B

rectification

A diode conducts in one direction only, so it can rectify AC. Amplification is the job of a transistor.

5. C

emitter

The emitter must supply many majority carriers. The base is thin and lightly doped; the collector is moderately doped and largest.

6. C

the emitter–base junction is forward biased and the collector–base junction is reverse biased

Input junction forward biased, output junction reverse biased: the active region.

7. **I_B = 0.05 mA; α = 0.95; β = 19**

$I_B = I_E - I_C$; $\alpha = I_C/I_E$; $\beta = I_C/I_B$.

8. **I_B = 0.20 mA; I_E = 30.2 mA**

$I_B = I_C/\beta = 30/150 = 0.20$ mA; $I_E = I_B + I_C = 30.2$ mA.

CHAPTER 6 · QUICK REVISION

Electromagnetism and Electronics — at a glance

CHAPTER 6 FORMULA SHEET

Topic	Formulas
Charge, Coulomb	$q = ne$; $F = k\lvert q_1 q_2\rvert / r^2$; $k = 8.99 \times 10^9$ N·m²/C²
Field, potential	$E = F/q = kq/r^2$; $V = kq/r$; $W = q\Delta V$; $E = V/d$; 1 eV $= 1.6 \times 10^{-19}$ J
Capacitors	$C = Q/V = \varepsilon_0 A/d$; $U = \frac{1}{2}CV^2$; parallel add; series add reciprocals
Current	$I = Q/t$; $V = IR$; $R = \rho L/A$; $P = IV = I^2R = V^2/R$
Circuits	series $R = \sum R_i$; parallel $1/R = \sum 1/R_i$; $V = \varepsilon - Ir$; Kirchhoff: $\sum I = 0$, $\sum \Delta V = 0$
Magnetism	$F = qvB\sin\theta$; $r = mv/qB$; $F = BIL\sin\theta$; $B = \mu_0 I/2\pi r$; $B = \mu_0 nI$
Flux, induction	$\Phi = BA\cos\theta$; $\varepsilon = -N\Delta\Phi/\Delta t$; $\varepsilon = BLv$; $V_s/V_p = N_s/N_p$
Transistor	$I_E = I_B + I_C$; $\alpha = I_C/I_E$; $\beta = I_C/I_B$

ELECTRICITY IN FIVE LINES

1. Like charges repel; force ∝ $q_1 q_2/r^2$.
2. Field points away from +, toward −; potential is a scalar.
3. Series: same current; parallel: same voltage.
4. Moving charges make and feel magnetic forces; a changing flux makes an emf.
5. Gap decides the material; doping makes n (donor, electrons) and p (acceptor, holes); diodes rectify, transistors amplify and switch.